

MAncor Method for Weir Management in the Grande Ronde and Imnaha Subbasins

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Goals and Purpose

This method is intended to provide a biologically grounded, adaptive approach for managing natural-origin and hatchery-origin fish at the weir. It is built within the same general management framework previously established by the comanagers and is not intended to change fishery management, harvest allocation, or comanager processes.

The method is designed to:

- Use observed biological metrics, long-term monitoring data, and recruitment relationships to inform management responses, and allow those inputs to be updated as new information becomes available rather than relying on fixed assumptions alone.
- Preserve fisheries within the general management framework previously established by the comanagers, with harvest impacts and allocation methods unchanged from prior agreements.
- Provide more proportional accounting for broodstock collection and natural spawning as abundance changes, supporting fuller seeding of available habitat when natural-origin abundance is low while avoiding excessive density-dependent effects when abundance is higher.
- Provide a durable framework suitable for long-term permitting under a 4(d) framework, with a structured path for incorporating improved science and updated biological observations through the comanager process over time.
- Improve implementation transparency by making management responses easier to trace back to observed biological conditions, while acknowledging forecast uncertainty, data quality limitations, and the need for comanager judgment in annual implementation.

Relationship to Prior Management

The Anchor Method builds from the same management foundation that has guided hatchery program implementation in the basin. The prior sliding scale approach was a reasonable framework given what was known at the time it was developed. It used stepped abundance categories and fixed thresholds to guide adult disposition, and those thresholds reflected the best available understanding of population abundance and biological requirements when they were established.

Since that time, the basin has accumulated substantial long-term monitoring data, improved abundance estimates, juvenile production information, and the ability to develop recruitment relationships that were not previously available. The Anchor Method is intended to make use of that accumulated information rather than continuing to apply assumptions developed before it existed.

A key structural difference is that the prior sliding scale applied management responses in steps, with the same response applied across an entire abundance range. This could produce meaningfully different outcomes for fish near the top versus the bottom of a given range, and created abrupt transitions when estimated abundance crossed a threshold. The Anchor Method applies management responses more continuously as abundance changes, which reduces the influence of arbitrary step boundaries and makes transitions smoother.

The following table summarizes the main structural differences between the two approaches.

Prior Sliding Scale	Anchor Method
Used stepped abundance categories to guide adult disposition.	Applies management responses more continuously as abundance changes.
Relied on abundance thresholds established when less biological information was available.	Uses observed biological metrics, monitoring data, and recruitment relationships that can be updated as methods and data improve.
Could produce abrupt changes when estimated abundance crossed a step threshold.	Reduces the influence of arbitrary step boundaries and provides smoother transitions.
Could treat fish near the high and low ends of an abundance range similarly.	Adjusts accounting more continuously so management responses better reflect current conditions.
Fixed threshold structure limited ability to incorporate new biological information.	Designed to accept improved inputs as field methods, analytical approaches, and biological understanding evolve.
Could require more natural-origin broodstock than desirable at the low end of a step.	Provides more proportional broodstock accounting tied to current abundance relative to the Anchor.

How Inputs Are Determined and Updated

A central design principle of the method is that inputs can be updated as better information becomes available. The method does not require any particular approach for deriving values like the spawner goal or the Wallowa/Lostine scaling factor. Instead,

it is designed to accept whatever values comanagers agree best represent current biological understanding, derived through whatever methods are most appropriate at the time.

This means that as field methods improve, as new analytical approaches become available, or as biological understanding of these populations develops, inputs can be revised through the comanager process without requiring structural changes to the accounting method itself. The accounting framework can remain consistent while the inputs it uses are refined over time.

For the current proof-of-concept version, some inputs are entered as working values pending comanager agreement on derivation methods. These are not permanent assumptions. They are placeholders that identify where comanager agreement is needed before the method is ready for permitting. The accounting structure does not depend on any particular value for these inputs — it depends only on comanagers agreeing on how those values will be determined and updated going forward.

Implementation Considerations

The method is intended to inform implementation, accounting, and review — not to replace comanager judgment, annual planning, or in-season management discretion. Comanagers will still need to evaluate current conditions, apply appropriate implementation buffers, consider operational constraints, and coordinate with permitting agencies as needed. The method improves transparency by making management responses easier to trace back to observed biological conditions, but that traceability should not be interpreted as implying that calculated values are targets to be matched exactly in every year.

The method is data-dependent. It is strongest where long-term monitoring, origin composition data, abundance estimates, juvenile production estimates, and recruitment relationships are available and reasonably well understood. The Lostine River provides a useful developmental example because it has this type of long-term information and an established weir-based management structure. Expansion to other populations should consider whether available data are sufficient to support the same type of biologically anchored accounting. As monitoring improves and analytical methods develop, inputs can be revised through the comanager process without requiring structural changes to the accounting framework.

Calculated accounting values are biologically informed estimates, not precise predictions of what will occur during a run year. Forecast uncertainty, in-season run timing, origin composition, fishery impacts, broodstock collection, and final escapement estimates all affect how closely implementation can match calculated values — even when in-season management is implemented appropriately. For this reason, some constraining metrics will need practical implementation buffers, acceptable ranges, or minimum and maximum bounds. For example, the method may identify a proposed pNOB value based on the best available preseason estimates. In season, managers implement broodstock collection toward that value, but final postseason pNOB will differ once actual returns, broodstock composition, fishery impacts, and escapement are fully estimated. A reasonable implementation structure would therefore include a planned

pNOB, a minimum acceptable pNOB, and an upper limit on the proportion or number of natural-origin fish that may be collected for broodstock. A minimum pNOB helps ensure enough natural-origin fish are incorporated into broodstock to meet conservation and integration expectations. A maximum limit prevents excessive removal of natural-origin fish from the spawning population. These buffers recognize normal implementation uncertainty without weakening the biological basis of the method.

Postseason estimates remain the final evaluation of what occurred and are important for assessing whether implementation was generally consistent with the method and for informing future adjustments. They should not, however, be interpreted as a perfect representation of what managers could have known or controlled during in-season implementation.

Lostine Management Area Accounting Toolkit

Purpose

This document explains the accounting process used to move from adult abundance estimates to broodstock allocation, fishery impacts, weir partitioning, spawner estimates, hatchery-origin disposition, and final pHOS, pNOB, and PNI values. It explains the management accounting logic and defines the major inputs and calculated terms used by the code. The goal is to make each value understandable to a reviewer: what it represents, why it matters, what caveats apply, and how it feeds the next accounting step.

A central conceptual point is that the Anchor is a separate natural-origin benchmark used only to calculate the pNOB slope. Once the Anchor and the scenario NO estimate are known, the remaining accounting proceeds stepwise through fisheries, weir partitioning, brood allocation, spawning, hatchery-origin placement, and final indices.

Accounting Overview

Part A — Set the Anchor

The Anchor is a natural-origin-only benchmark. It estimates the minimum natural-origin adult abundance required to satisfy both full broodstock need and the spawner goal using only natural-origin fish, after natural-origin fishery impacts are applied. The Anchor does not calculate final mixed-origin outcomes. It creates the reference abundance used to calculate the pNOB slope in Part B.

Part B — Apply the Anchor to Scenario Accounting

Part B applies the Anchor to the actual or scenario abundance estimates. The management priority order in Part B is:

- Protect broodstock
- Allow fisheries from remaining available fish

- Allow spawning with fish remaining after broodstock and fisheries
 - Remove excess captured hatchery-origin fish
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Part A — The Anchor

anchor_required_no

Definition

`anchor_required_no` is the minimum natural-origin adult abundance needed in the management area to fully satisfy the method's natural-origin objectives under the current assumptions. It asks:

How many natural-origin fish would need to be present so that, after expected NO fishery impacts occur, enough NO fish remain to both fill the entire broodstock need and meet the adult spawner goal using only NO fish?

The Anchor is calculated by testing candidate NO abundance values. For each candidate, the accounting applies NO sport and treaty impacts, estimates remaining NO fish, partitions those fish at the weir, allocates brood entirely from NO fish, applies pre-spawn survival, and checks whether both broodstock need and the spawner goal are met. The broodstock feasibility check compares brood need directly to NO fish captured at the weir before survival is applied, while the spawner goal check uses survival-adjusted NO spawners. The Anchor is set at the lowest candidate abundance that satisfies both requirements.

The Anchor is not the final accounting result for the year. It does not include HO fish, pHOS, final pNOB, PNI, or final HO placement. In Part B it is used only to calculate the pNOB slope.

Relationship to pNOB slope

$$\text{pnob_slope} = \min(1, \text{no_manarea_est} / \text{anchor_required_no})$$
$$\text{no_brood_desired} = \text{pnob_slope} \times \text{brood_need}$$

From those two values the remaining accounting proceeds stepwise: fishery impacts and harvest are applied subject to broodstock protection, fish are partitioned at the weir, broodstock is allocated, spawners are estimated after survival, captured HO fish are placed above the weir only if needed to meet the spawner goal, and excess captured HO fish are removed.

Part B — Scenario Accounting

Part B applies the Anchor to actual or scenario abundance estimates. The priority order (broodstock first, then fisheries, then spawning, then HO removal) determines the sequence of every accounting step that follows.

The broodstock protection cap is the most conceptually unusual part of the method and worth understanding before reading the glossary entries. Preliminary fishery impacts and harvest are calculated first using the applicable curves and harvest-sharing framework. Those preliminary values are then checked against the broodstock protection cap: if projected impacts would leave too few fish available to meet the NO brood target after accounting for the expected trap proportion, sport and treaty impacts are scaled down proportionally until the cap is satisfied. The same logic applies to HO harvest if projected HO harvest would leave too few HO fish to fill any remaining brood need not met by NO fish. These caps protect broodstock availability only, they do not reserve fish for spawning ahead of fisheries, and they do not change the underlying fishery calculation methods.

The HO placement decision is the other conditional step worth understanding before the glossary. After broodstock and fisheries are accounted for and spawners are estimated from naturally distributed NO and HO fish, the accounting checks whether the spawner goal is met. If a deficit remains, captured HO fish available after broodstock are placed above the weir, up to the number needed to close the deficit after above-weir survival is applied. Any captured HO remaining after broodstock and necessary spawning placement are removed.

Glossaries

How to Use This Glossary

This glossary is intended to be used alongside the accounting output tables. Reviewers do not need to read every entry in order. The input glossary explains where values come from; the calculated term glossary explains how values move through the accounting.

How Inputs Are Selected

use_csv_scenario_values = FALSE: Scenario values entered directly in the code are used. This is the normal approach for planning, proof-of-concept testing, or evaluating alternative scenarios.

use_csv_scenario_values = TRUE: The tool pulls scenario values from the selected CSV year identified by `csv_scenario_year`. For NO and HO abundance estimates, the tool uses the best available estimate in priority order: post-season estimate first, then in-season estimate, then pre-season estimate.

For accounting parameters such as `trap_prop_no`, `trap_prop_ho`, `weir_efficiency`, `survival_above_weir`, and `survival_below_weir`, the tool uses the selected year's actual value when available. If the selected year's value is missing, the tool falls back to a

median calculated from the most recent valid non-missing years in the CSV. The code requires at least three valid years for each median-derived parameter, meaning different parameters may use different sets of years depending on data availability.

Some values remain user-entered management assumptions even when CSV mode is used. `spawner_goal` and `wl_scaling` are currently entered directly unless future versions include agreed methods and data fields for deriving them automatically. These values should be revisited over time as part of adaptive management when improved methods or co-manager-approved input sources are developed.

Input Data Requirements

The following table summarizes each input, its source, and whether a median fallback is available.

Input	CSV Column	Required	Median Fallback	Default
<code>no_manarea_est</code>	Pre.NO / In.NO / Post.NO	Yes	No	Manual entry
<code>ho_manarea_est</code>	Pre.HO / In.HO / Post.HO	Yes	No	Manual entry
<code>brood_need</code>	BS.Need	Yes	No	Manual entry
<code>spawner_goal</code>	— (manual)	Yes	No	Manual entry
<code>wl_scaling</code>	— (manual)	Yes	No	Manual entry
<code>trap_prop_no</code>	Trap.NO / Post.NO	Yes	Yes	5-yr median
<code>trap_prop_ho</code>	Trap.HO / Post.HO	Yes	Yes	5-yr median
<code>weir_efficiency</code>	Weir.Eff	Yes	Yes	5-yr median
<code>survival_above_weir</code>	PSS.ABV	Yes	Yes	5-yr median
<code>survival_below_weir</code>	PSS.BLW	Yes	Yes	5-yr median
<code>utilization_no</code>	— (manual)	No	No	1.0
<code>utilization_ho</code>	— (manual)	No	No	1.0

Input Glossary

Stock

Definition

The population or stock identifier for the row of data.

How it is used

The accounting tool filters the CSV to use Lostine stock rows only. In the current tool, the relevant stock value is LST.

Source

CSV column: Stock

Year

Definition

The management year associated with the row of data.

How it is used

When `use_csv_scenario_values = TRUE`, `csv_scenario_year` identifies which Year row should be used for scenario values and selected-year accounting parameters.

Source

CSV column: Year

no_manarea_est / Pre.NO, In.NO, Post.NO

Definition

The estimated natural-origin adult abundance in the Lostine management area for the scenario being evaluated.

Notes

This is the starting NO abundance for Part B scenario accounting. It is compared to `anchor_required_no` to calculate the pNOB slope. When CSV scenario mode is used, the tool uses the best available NO estimate in priority order: `Post.NO` → `In.NO` → `Pre.NO`.

Source

When `use_csv_scenario_values = FALSE`, user-entered scenario value. When `use_csv_scenario_values = TRUE`, pulled from the selected CSV year using the estimate priority described above.

ho_manarea_est / Pre.HO, In.HO, Post.HO

Definition

The estimated hatchery-origin adult abundance in the Lostine management area for the scenario being evaluated.

Notes

This value is used in Part B only. HO fish are not used to set the Anchor. HO fish may contribute to broodstock, fisheries, and natural spawning depending on availability after

the priority order is applied. When CSV scenario mode is used, the tool uses the best available HO estimate in priority order: Post.HO → In.HO → Pre.HO.

Source

When `use_csv_scenario_values = FALSE`, user-entered scenario value. When `use_csv_scenario_values = TRUE`, pulled from the selected CSV year using the estimate priority described above.

brood_need / BS.Need

Definition

The total adult broodstock need for the scenario.

Notes

Broodstock is the first priority in Part B. The pNOB slope determines the desired NO portion of broodstock, and any remaining brood need may be filled with HO fish if available. The brood need is determined through the AOP process and is based on fecundity assumptions and egg collection goals.

Source

When `use_csv_scenario_values = FALSE`, user-entered scenario value. When `use_csv_scenario_values = TRUE`, pulled from the selected CSV year using the BS.Need column.

spawner_goal

Definition

The adult spawner goal is the target number of adults intended to optimize natural production of offspring in the Lostine River. It is used to evaluate whether enough adults are expected to survive to spawning after fishery impacts, broodstock collection, weir partitioning, and pre-spawn mortality are accounted for.

Notes

The spawner goal is used in both Part A and Part B. In Part A it is part of the Anchor feasibility test. In Part B it determines whether captured HO fish should be placed above the weir after broodstock and fisheries have been accounted for. For this proof-of-concept version the spawner goal is based on visual review of existing recruitment curves and is not a final derived management target. The intended derivation is a spawner-recruit analysis using the long-term Lostine monitoring dataset. A Beverton-Holt or Ricker recruitment curve fitted to observed spawner and recruit pairs would identify the spawner abundance that maximizes recruits per spawner or achieves a specified fraction of maximum recruitment. The data to perform this analysis exist in the current monitoring record; the remaining step is co-manager agreement on the curve form and the specific target point on that curve.

Current value

`spawner_goal = 800` (proof-of-concept placeholder)

wl_scaling

Definition

The Wallowa/Lostine scaling factor used to expand the Lostine population segment to the full Wallowa/Lostine population scale used by the fishery impact curves.

Notes

Sport and treaty curves use the Wallowa/Lostine-scaled abundance. Sport NO impacts are converted back to Lostine-only impacts before being subtracted from Lostine NO abundance because some NO impacts in the sport fishery may not be fish returning to the Lostine River. This method assumes proportionality. Treaty NO impacts are calculated from Wallowa/Lostine-scaled abundance but are not scaled back before being subtracted from Lostine NO abundance because all treaty harvest generally occurs within the Lostine River itself. The intended derivation is a multi-year median of the ratio of Lostine redds to total Wallowa/Lostine redds from spawning ground survey data. New methods for estimating population distribution should be considered if co-managers agree they provide improved estimates. The current value of 1.4 reflects the historically applied expansion factor and should be formally recalculated from recent redd distribution data before permitting.

Current value

wl_scaling = 1.4 (historically applied expansion factor)

trap_prop_no / Trap.NO, Post.NO

Definition

The estimated proportion of NO adults captured at the weir after fisheries have occurred.

Notes

This value affects how many NO fish are available for broodstock and how many are estimated to have passed above the weir uncaptured.

Calculation

$$\text{trap_prop_no} = \text{Trap.NO} / \text{Post.NO}$$

Source

Selected year actual value when available; otherwise median fallback from most recent valid years.

trap_prop_ho / Trap.HO, Post.HO

Definition

The estimated proportion of HO adults captured at the weir after fisheries have occurred.

Notes

This value affects how many HO fish are available for broodstock, possible placement above the weir, or removal.

Calculation

$$\text{trap_prop_ho} = \text{Trap.HO} / \text{Post.HO}$$

Source

Selected year actual value when available; otherwise median fallback from most recent valid years.

weir_efficiency / Weir.Eff

Definition

The estimated annual weir capture efficiency used in the accounting to estimate fish that passed above the weir without being captured.

Notes

Provided by Fish Research and derived from spawning ground observations above the weir. Fish captured at the weir are uniquely marked. During carcass surveys above the weir, marked carcasses divided by total inspected carcasses gives the annual weir capture efficiency. The accounting applies weir_efficiency equally to NO and HO fish. The value can vary by management year depending on river conditions, timing, weir performance, and operational constraints.

Use

$$\text{uncaptured_above} = (\text{captured} / \text{weir_efficiency}) - \text{captured}$$

Source

Selected year actual value when available; otherwise median fallback from most recent valid years.

survival_above_weir / PSS.ABV

Definition

The pre-spawn survival rate applied to fish located above the weir.

Notes

Fish above the weir are not automatically counted as spawners. Survival must be applied before they become spawners in the accounting.

Use

$$\text{spawners_above} = \text{fish_above} \times \text{survival_above_weir}$$

Source

Selected year actual value when available; otherwise median fallback from most recent valid years.

survival_below_weir / PSS.BLW

Definition

The pre-spawn survival rate applied to fish remaining below the weir.

Notes

Fish below the weir are not automatically counted as spawners. Survival must be applied before they become spawners in the accounting.

Use
$$\text{spawners_below} = \text{fish_below} \times \text{survival_below_weir}$$
Source

Selected year actual value when available; otherwise median fallback from most recent valid years.

utilization_no, utilization_ho

Definition

The proportion of allowed NO impacts and HO harvest actually taken in the fishery. At the default value of 1.0, full utilization is assumed.

Notes

Reducing these values models years when fisheries do not fully take the allowed impacts or harvest, leaving more fish in the system post-fishery. Utilization is applied after broodstock-protection caps, so broodstock availability is always calculated at full allowed impact regardless of utilization.

Default

utilization_no = 1.0, utilization_ho = 1.0

Calculated Term Glossary**anchor_required_no**

Definition

The minimum NO adult abundance needed to meet the Anchor objectives under the current assumptions. The Anchor is a NO-only benchmark that asks how many NO fish are needed so that, after NO fishery impacts, enough NO fish remain to satisfy broodstock need and the spawner goal using only NO fish.

Notes

The Anchor is not the final mixed-origin accounting result. It does not include HO fish, pHOS, final pNOB, PNI, or final HO placement. In Part B, anchor_required_no is used to calculate the pNOB slope. The broodstock feasibility check compares brood need directly to NO fish captured at the weir before survival is applied, while the spawner goal check uses survival-adjusted NO spawners.

Calculation

Produced through the Part A Anchor process. The accounting tests candidate NO abundances and selects the lowest candidate that satisfies both broodstock need and the spawner goal after NO fishery impacts.

pnob_slope

Definition

The target proportion of broodstock that should be natural-origin, based on how the scenario NO abundance compares to the Anchor.

Notes

This is a target value used to set the desired NO brood contribution. It is not the final realized pNOB. Final realized pNOB is calculated later from actual brood allocation.

Calculation

$$\text{pnob_slope} = \min(1, \text{no_manarea_est} / \text{anchor_required_no})$$

no_wl

Definition

The scenario NO abundance scaled to the full Wallowa/Lostine population basis used by fishery curves. This is not the number of Lostine NO fish — it is an expanded value used for fishery impact calculations.

Calculation

$$\text{no_wl} = \text{wl_scaling} \times \text{no_manarea_est}$$

sport_impact_wl_projected

Definition

The preliminary sport fishery NO impact calculated from the sport fishery curve on the Wallowa/Lostine scale before broodstock-protection caps are applied.

Notes

This is a projected curve value, not necessarily the final allowed sport impact. It is calculated on the Wallowa/Lostine scale and converted back to Lostine scale before it is used as a Lostine NO impact.

Calculation

$$\text{sport_impact_wl_projected} = \text{sport_curve}(\text{no_wl})$$

sport_no_impacts_projected

Definition

The preliminary sport fishery impact on Lostine NO fish after converting the Wallowa/Lostine-scale sport impact back to the Lostine scale.

Notes

This value may be capped to protect broodstock. The capped value is sport_no_impacts.

Calculation

$$\text{sport_no_impacts_projected} = \text{sport_impact_wl_projected} / \text{wl_scaling}$$

treaty_no_impacts_projected

Definition

The preliminary treaty fishery impact on NO fish calculated from Wallowa/Lostine-scaled abundance.

Notes

Unlike sport NO impacts, treaty NO impacts are not scaled back to Lostine. Treaty impacts occur within the Lostine River itself, while sport impacts occur in the Wallowa River where fish from outside the Lostine are also encountered. Treaty NO impacts are subtracted directly from Lostine NO abundance by management convention.

Calculation

$$\text{treaty_no_impacts_projected} = \text{treaty_curve}(\text{no_wl})$$

total_no_impacts_projected

Definition

The total preliminary NO fishery impacts before broodstock-protection caps are applied.

Calculation

$$\text{total_no_impacts_projected} = \text{sport_no_impacts_projected} + \text{treaty_no_impacts_projected}$$

no_brood_target

Definition

The desired number of NO fish in broodstock based on the pNOB slope.

Notes

This is the NO brood target used when applying the broodstock-protection cap to NO fisheries. It is not necessarily the final actual NO brood collected, because actual collection is limited by NO fish captured at the weir.

Calculation

$$\text{no_brood_target} = \text{pnob_slope} \times \text{brood_need}$$

required_no_after_fisheries

Definition

The estimated NO abundance that must remain after fisheries so enough NO fish can be captured at the weir to meet the NO brood target.

Notes

This value is used only to protect broodstock availability. It does not reserve NO fish for spawning ahead of fisheries.

Calculation

```
required_no_after_fisheries = no_brood_target / trap_prop_no
```

max_no_impacts_allowed

Definition

The maximum total NO fishery impacts allowed while still leaving enough NO abundance after fisheries to meet the NO brood target, based on the NO trap proportion.

Notes

This cap protects broodstock only. It does not reserve NO fish for spawning ahead of fisheries.

Calculation

```
required_no_after_fisheries = no_brood_target / trap_prop_no  
max_no_impacts_allowed = max(0, no_manarea_est -  
    required_no_after_fisheries)
```

sport_no_impacts and treaty_no_impacts

Definition

The final allowed sport and treaty NO impacts after applying the broodstock-protection cap.

Notes

If projected NO impacts are less than or equal to max_no_impacts_allowed, these values equal the projected values. If projected NO impacts exceed the cap, sport and treaty impacts are scaled down proportionally so their sum equals the allowed total.

Calculation

```
c(sport_no_impacts, treaty_no_impacts) =  
scale_to_cap(c(sport_no_impacts_projected,  
    treaty_no_impacts_projected), max_no_impacts_allowed)
```

total_no_impacts_allowed

Definition

The total allowed NO fishery impact after broodstock-protection caps are applied. May be less than total_no_impacts_projected if projected impacts would leave too few NO fish to meet the NO brood target.

Calculation

```
total_no_impacts_allowed = sport_no_impacts + treaty_no_impacts
```

scale_to_cap()

Definition

A helper calculation that reduces projected fishery impacts or harvest proportionally when the projected total exceeds the broodstock-protection cap.

Notes

This preserves the projected split between fisheries while reducing the total to the allowed maximum. If the projected total is already below the cap, values are unchanged. Values are constrained so impacts and harvest do not become negative.

Calculation

```
scaled_values = projected_values × (cap / projected_total)
```

ho_sport_harvest_projected

Definition

The preliminary HO sport harvest estimate before the HO broodstock-protection cap is applied.

Notes

Derived from allowed Lostine NO handling and the HO:NO abundance ratio. The sport fishery targets adipose-clipped hatchery fish; wild (NO) fish are encountered incidentally and released. The FMEP (section 1.4.3) establishes 10% as the assumed catch-and-release hooking mortality rate. Working backwards from allowed NO mortality gives total NO fish that can be handled, and scaling by the HO:NO ratio gives expected HO harvest. This approach anchors HO sport harvest to the allowed NO impact rather than running a separate HO harvest curve, keeping the method internally consistent with the existing FMEP fishery framework. This value may be reduced if HO harvest must be capped to protect broodstock.

Calculation

```
allowed_no_handle = 10 × sport_no_impacts  
ho_sport_harvest_projected = allowed_no_handle × (ho_manarea_est /  
no_manarea_est)
```

ho_treaty_harvest_projected

Definition

The preliminary HO treaty harvest estimate before the HO broodstock-protection cap is applied.

Notes

HO treaty harvest is calculated as a catch-balancing residual, not from a separate HO treaty curve. The co-manager fishery framework operates on a 50/50 total harvest share between tribal and non-tribal fisheries (FMEP section 2.1.3, Table 7). The sport fishery generates NO incidental mortality and HO harvest together. To balance the allocation,

the treaty side receives HO fish equal to the residual after their NO impacts are subtracted from the total sport fishery value.

Calculation

```
ho_treaty_harvest_projected = max(0, (sport_impact_wl +  
ho_sport_harvest_projected) - treaty_no_impacts)
```

Worked example

Using default scenario values (NO = 382, HO = 536, wl_scaling = 1.4): no_wl = 534.8. The sport curve returns 7 WL-scale NO fish, converting to 5 Lostine-scale NO fish ($7 \div 1.4$). The treaty curve returns 22 NO fish applied directly at Lostine scale. Allowed NO handling = $5 \div 0.10 = 50$ fish. With HO:NO ratio = 1.40, HO sport harvest = 70 fish. Total sport fishery = $7 + 70 = 77$. Tribal fishery has already taken 22 NO fish, so HO treaty harvest = $77 - 22 = 55$. As a check: tribal receives 22 NO + 55 HO = 77 total, matching the non-tribal sport total of 77. This result is floored at zero.

ho_brood_needed

Definition

The amount of broodstock need remaining after the estimated NO brood contribution is accounted for.

Notes

Used to determine how many HO fish must remain available after fisheries to protect broodstock needs. If NO brood is sufficient to meet the brood need, then ho_brood_needed is zero.

Calculation

```
ho_brood_needed = max(0, brood_need -  
no_brood_actual_for_brood_check)
```

required_ho_after_fisheries

Definition

The estimated HO abundance that must remain after fisheries so enough HO fish can be captured at the weir to fill the remaining brood need not met by NO fish.

Notes

Used only to protect broodstock availability. Does not reserve HO fish for spawning ahead of fisheries.

Calculation

```
required_ho_after_fisheries = ho_brood_needed / trap_prop_ho
```

max_ho_harvest_allowed

Definition

The maximum total HO harvest allowed while still leaving enough HO abundance after fisheries to fill any brood need not met by NO fish.

Notes

This cap protects broodstock only. HO fish remaining after broodstock and fisheries may contribute to spawning if needed. Excess captured HO fish are removed.

Calculation

```
required_ho_after_fisheries = ho_brood_needed / trap_prop_ho  
max_ho_harvest_allowed = max(0, ho_manarea_est -  
required_ho_after_fisheries)
```

ho_sport_harvest and ho_treaty_harvest

Definition

The final allowed HO sport and treaty harvest after applying the HO broodstock-protection cap.

Notes

If projected HO harvest is less than or equal to max_ho_harvest_allowed, these values equal the projected values. If projected HO harvest exceeds the cap, HO sport and treaty harvest are scaled down proportionally so their sum equals the allowed total.

Calculation

```
c(ho_sport_harvest, ho_treaty_harvest) =  
scale_to_cap(c(ho_sport_harvest_projected,  
ho_treaty_harvest_projected), max_ho_harvest_allowed)
```

total_ho_harvest_allowed

Definition

The total allowed HO harvest after broodstock-protection caps are applied. May be less than total_ho_harvest_projected if projected HO harvest would leave too few HO fish available to fill brood need not met by NO fish.

Calculation

```
total_ho_harvest_allowed = ho_sport_harvest + ho_treaty_harvest
```

no_after_fisheries and ho_after_fisheries

Definition

The estimated NO and HO adults remaining after allowed fishery impacts or harvest are applied. These are the post-fishery abundance values that become the starting point for weir partitioning.

Notes

Utilization rates utilization_no and utilization_ho are applied here. At the default value of 1.0, full utilization of allowed impacts and harvest is assumed. Reducing these values

models years when fisheries do not fully take the allowed impacts or harvest, leaving more fish in the system post-fishery.

Calculation

```
no_actual_impacts = (sport_no_impacts + treaty_no_impacts) ×  
utilization_no  
ho_actual_harvest = (ho_sport_harvest + ho_treaty_harvest) ×  
utilization_ho  
no_after_fisheries = no_manarea_est - no_actual_impacts  
ho_after_fisheries = ho_manarea_est - ho_actual_harvest
```

no_captured_weir and ho_captured_weir

Definition

The estimated number of NO and HO adults captured at the weir after fisheries.

Notes

Captured fish are available for broodstock first. Captured NO not used for brood are treated as available above the weir. Captured HO not used for brood may be placed above the weir if needed for the spawner goal, otherwise removed.

Calculation

```
no_captured_weir = trap_prop_no × no_after_fisheries  
ho_captured_weir = trap_prop_ho × ho_after_fisheries
```

no_above_weir_uncaptured and ho_above_weir_uncaptured

Definition

The estimated number of adult fish that were not captured and escaped above the weir before pre-spawn mortality is applied.

Notes

Fish may pass the weir without being captured by passing prior to weir installation, through gaps in the weir, or when the weir is temporarily raised or lowered for maintenance. These fish are not yet counted as spawners; above-weir survival must be applied first.

Calculation

```
no_above_weir_uncaptured = (no_captured_weir / weir_efficiency) -  
no_captured_weir  
ho_above_weir_uncaptured = (ho_captured_weir / weir_efficiency) -  
ho_captured_weir
```

no_below_weir and ho_below_weir

Definition

The estimated number of adult fish remaining below the weir after accounting for captured fish and uncaptured fish that passed above the weir. These are often fish that never arrive at the weir.

Notes

These fish are not yet counted as spawners. Below-weir survival must be applied first.

Calculation

```
no_below_weir = no_after_fisheries - (no_captured_weir +  
no_above_weir_uncaptured)  
ho_below_weir = ho_after_fisheries - (ho_captured_weir +  
ho_above_weir_uncaptured)
```

no_brood_desired

Definition

The desired NO contribution to broodstock based on the pNOB slope.

Notes

This is a target, not necessarily the actual number of NO fish collected for broodstock.

Calculation

```
no_brood_desired = pnob_slope × brood_need
```

no_brood_actual and ho_brood_actual

Definition

The final estimated broodstock allocation by origin.

Notes

NO brood is allocated first up to no_brood_desired. HO brood is then used to fill any remaining broodstock need if captured HO fish are available.

Calculation

```
no_brood_actual = min(no_brood_desired, no_captured_weir)  
ho_brood_actual = min(brood_need - no_brood_actual,  
ho_captured_weir)
```

no_captured_available_above and ho_captured_available_above

Definition

Captured fish remaining after broodstock allocation.

Notes

Captured NO remaining after broodstock are treated as available above the weir.

Captured HO remaining after broodstock may be placed above the weir only if needed to meet the spawner goal; otherwise they are removed.

Calculation

```
no_captured_available_above = no_captured_weir - no_brood_actual  
ho_captured_available_above = ho_captured_weir - ho_brood_actual
```

no_spawnners_above and no_spawnners_below

Definition

The estimated NO adults surviving to spawn above and below the weir.

Notes

The word spawners indicates that pre-spawn survival has already been applied.

Calculation

```
no_spawnners_above = (no_above_weir_uncaptured +  
no_captured_available_above) × survival_above_weir  
no_spawnners_below = no_below_weir × survival_below_weir
```

ho_spawnners_above_uncapt and ho_spawnners_below

Definition

The estimated HO adults surviving to spawn above and below the weir before any additional captured HO placement above the weir.

Notes

ho_spawnners_above_uncapt includes HO fish that passed above the weir without being captured and then survived to spawning. It does not include captured HO fish intentionally placed above the weir.

Calculation

```
ho_spawnners_above_uncapt = ho_above_weir_uncaptured ×  
survival_above_weir  
ho_spawnners_below = ho_below_weir × survival_below_weir
```

current_spawnners_without_HO_captured_above

Definition

The estimated total system spawners before placing any additional captured HO fish above the weir.

Notes

Used to determine whether captured HO fish are needed above the weir to meet the spawner goal after broodstock and fisheries have already occurred.

Calculation

```
current_spawnners_without_HO_captured_above = no_spawnners_above + no_  
spawnners_below + ho_spawnners_above_uncapt + ho_spawnners_below
```

ho_captured_above

Definition

The estimated number of captured HO fish placed above the weir before pre-spawn survival is applied.

Notes

Captured HO fish are placed above the weir only if needed to meet the spawner goal. They are not automatically counted as spawners; above-weir survival is applied before they contribute to `ho_spawnners_above`.

Calculation

```
spawner_deficit = max(0, spawner_goal -  
current_spawnners_without_HO_captured_above)  
ho_needed_pre_survival = spawner_deficit / survival_above_weir  
ho_captured_above = min(ho_captured_available_above,  
ho_needed_pre_survival)
```

`ho_captured_removed`

Definition

The estimated number of captured HO fish removed after broodstock allocation and any needed HO placement above the weir.

Notes

Removal occurs after broodstock, fisheries, and needed spawning placement have been accounted for.

Calculation

```
ho_captured_removed = max(0, ho_captured_available_above -  
ho_captured_above)
```

`no_spawnners_total`, `ho_spawnners_total`, and `system_spawnners_total`

Definition

Final estimated adult spawner totals by origin and combined system total.

Notes

These values include survival adjustments. HO spawners include uncaptured HO spawners and any captured HO placed above the weir that survive to spawning.

Calculation

```
ho_spawnners_above = ho_spawnners_above_uncapt + (ho_captured_above ×  
survival_above_weir)  
no_spawnners_total = no_spawnners_above + no_spawnners_below  
ho_spawnners_total = ho_spawnners_above + ho_spawnners_below  
system_spawnners_total = no_spawnners_total + ho_spawnners_total
```

`phos`

Definition

The proportion of hatchery-origin spawners in the final estimated spawning population.

Notes

Based on estimated spawners after survival, not simply fish present above or below the weir before survival.

Calculation

$$\text{phos} = \text{ho_spawners_total} / \text{system_spawners_total}$$

pnob

Definition

The final proportion of natural-origin fish in broodstock.

Notes

This is the final realized broodstock composition, not just the desired pNOB slope.

Calculation

$$\text{pnob} = \text{no_brood_actual} / \text{brood_need}$$

pni

Definition

The proportionate natural influence index calculated from final pNOB and pHOS values.

Notes

PNI depends on both broodstock composition and spawning composition.

Calculation

$$\text{pni} = \text{pnob} / (\text{pnob} + \text{phos})$$